

Supplementary Material: Adaptive Action Updates in Offline Reinforcement Learning for Integrated Energy-Efficient Building Control

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A Offline Dataset Construction Details

This appendix provides the complete exploration, action-repetition, transition-storage, and preprocessing details for the offline dataset described in Section 4.4 of the main paper.

A.1 Subsystem-Specific Expert Exploration

Let the nominal expert action be

$$a_t^E = \mu_E(o_t),$$

where μ_E denotes the expert controller. The exploratory proposal used in episode e is

$$\tilde{a}_t^E = \mathcal{N}_{\eta_e}(a_t^E), \quad (1)$$

where $\mathcal{N}_{\eta_e}$ is a subsystem-specific perturbation operator and η_e identifies the exploration profile. Four profiles were used: clean, low, medium, and high.

A.1.1 Glazing Exploration. For each glazing control i , the nominal expert state is replaced by a randomly selected admissible state with probability ϵ_g :

$$\tilde{g}_{t,i}^E = \begin{cases} g_{t,i}^{\text{rand}}, & \text{with probability } \epsilon_g, \\ g_{t,i}^E, & \text{with probability } 1 - \epsilon_g, \end{cases} \quad i \in \{1, \dots, 4\}, \quad (2)$$

where $g_{t,i}^{\text{rand}}$ is sampled from the four admissible glazing states.

A.1.2 Lighting Exploration. For lighting zone z , Gaussian noise is applied with probability p_ℓ^{noise} :

$$\tilde{\ell}_{t,z}^E = \begin{cases} \text{clip}(\ell_{t,z}^E + \epsilon_{t,z}^\ell, 0, \ell_z^{\text{max}}), & \text{with probability } p_\ell^{\text{noise}}, \\ \ell_{t,z}^E, & \text{otherwise,} \end{cases} \quad (3)$$

where

$$\epsilon_{t,z}^\ell \sim \mathcal{N}(0, (\sigma_\ell \ell_z^{\text{max}})^2). \quad (4)$$

Scaling the standard deviation by ℓ_z^{max} accounts for differences in installed lighting power among zones.

The glazing and lighting exploration parameters are reported in Table S1.

Table S1: Glazing and lighting action-value exploration parameters.

Profile	ϵ_g	p_ℓ^{noise}	σ_ℓ
Clean	0.00	0.00	0.00
Low	0.05	0.15	0.03
Medium	0.10	0.30	0.05
High	0.20	0.40	0.08

A.1.3 Thermal Exploration. Thermal exploration depends on whether the zone air temperature lies inside or outside the occupied comfort band. Inside the band, the expert probabilistically selects among three modes: an idle command, a randomly sampled admissible set-point, and the active expert target. Outside the band, the corrective expert target is optionally perturbed by zero-mean Gaussian noise.

The corresponding parameters are reported in Table S2. For each profile, $p_{\text{idle}} + p_{\text{rand}} + p_{\text{target}} = 1$.

Here, $p_{\text{out}}^{\text{noise}}$ is the probability of perturbing a corrective target when the temperature lies outside the comfort band, and σ_T is the standard deviation of the temperature perturbation.

A.2 Probabilistic Action Repetition

Action-value exploration determines the candidate action $\tilde{a}_t^E$, whereas temporal exploration independently determines whether each candidate component is applied.

Let $q(i)$ identify the subsystem associated with action dimension i :

$$q(i) \in \{\text{glazing, lighting, thermal}\}.$$

A repetition indicator is sampled independently for each dimension:

$$\begin{aligned} \kappa_{t,i} &\sim \text{Bernoulli}(p_{q(i),e}^{\text{rep}}), \\ b_{t,i} &= 1 - \kappa_{t,i}, \end{aligned} \quad (5)$$

where $\kappa_{t,i} = 1$ denotes retention and $b_{t,i} = 1$ denotes an update. The executed action is consequently constructed according to Equation (10) of the main paper. This $b_{t,i}$ describes only the forced repetition; the update mask stored in the dataset and used for training is recomputed from the executed actions (Equation (11) of the main paper), so it is also zero when the expert itself reproduces its previous value.

The base repetition configuration was

$$\begin{aligned} p_{\text{glazing}}^{\text{rep}} &= 0.05, \\ p_{\text{lighting}}^{\text{rep}} &= 0.05, \\ p_{\text{thermal}}^{\text{rep}} &= 0.10. \end{aligned} \quad (6)$$



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Table S2: Thermal action-value exploration parameters.

Profile	p_{idle}	p_{rand}	p_{target}	$p_{\text{out}}^{\text{noise}}$	σ_T [°C]
Clean	1.00	0.00	0.00	0.00	0.00
Low	0.85	0.10	0.05	0.10	0.25
Medium	0.70	0.20	0.10	0.20	0.50
High	0.55	0.30	0.15	0.30	0.70

Additional episodes used stronger repetition probabilities to provide examples of longer action-holding durations. The action-value exploration profile and repetition configuration were assigned independently.

A.3 Episode Composition and Reproducibility

The pooled dataset contains 14 complete annual EnergyPlus episodes. Each episode contains 52,559 transitions at an approximately ten-minute control resolution. The episode assignments are

$$\begin{aligned}
 \mathcal{E}_{\text{clean}} &= \{000\}, \\
 \mathcal{E}_{\text{low}} &= \{001, 002\}, \\
 \mathcal{E}_{\text{medium}} &= \{003, \dots, 010, 012\}, \\
 \mathcal{E}_{\text{high}} &= \{011, 013\}.
 \end{aligned} \tag{7}$$

A deterministic episode-specific exploration seed was used:

$$\zeta_e = 10,000 + 1,000I_E + I_e, \tag{8}$$

where I_E and I_e denote the expert and episode identifiers, respectively.

A.4 Transition Representation

For each timestep, the stored transition is

$$\mathcal{T}_t = (o_t, a_{t-1}^{\text{exec}}, a_t^{\text{exec}}, b_t, c_t, r_t, o_{t+1}, d_t, e_t), \tag{9}$$

where d_t denotes episode termination and e_t identifies the source episode and exploration profile.

The stored proposal-conditioning vector is

$$c_t = (1 - b_t) \odot a_{t-1}^{\text{exec}} + b_t \odot \xi, \quad \xi = -2 \cdot \mathbf{1}. \tag{10}$$

All normalized physical actions lie in $[-1, 1]$; therefore, $\xi = -2 \cdot \mathbf{1}$ is an out-of-range marker. For a retained dimension, $c_{t,i}$ contains the previous executed value. For an updated dimension, it contains the marker value. Although c_t is stored in the dataset, it is deterministic given a_{t-1}^{exec} and b_t .

Annual episodes remain contiguous in the pooled dataset. The terminal indicator is set only for the final transition of each episode, producing 14 terminal transitions. No artificial daily or weekly episode boundaries are introduced, and recurrent training sequences are not permitted to cross annual episode boundaries.

A.5 Shared Preprocessing Statistics

Observation normalization follows Section 4.1 of the main paper. The means and standard deviations of the 23 physical observation variables are computed from all 735,826 transitions in the pooled offline training dataset. The six bounded temporal and occupancy variables, corresponding to dimensions 24 through 29, are passed through without standardization.

Table S3: State-independent selectors at different update budgets (same checkpoint and proposals). $k \times$ learned: k times the learned per-subsystem update rates; RBC rate: the behaviour controller’s own per-subsystem rates. Means over selector seeds.

Selector	Budget	kWh/d	Viol. %	Kzh/d	Vis. %	Upd. %
Learned (sampled)	–	344.1	7.4	1.23	48.4	13.7
Periodic	1× learned	301.6	22.0	5.79	27.7	13.6
	2× learned	328.5	18.9	5.07	48.6	27.2
	4× learned	363.1	14.6	3.18	58.7	54.4
	RBC rate	339.8	17.8	4.68	29.1	26.9
Random	1× learned	296.9	23.9	6.48	20.9	13.6
	2× learned	314.4	19.8	5.11	33.3	27.2
	4× learned	357.8	14.7	3.03	48.9	54.3
	RBC rate	313.9	19.6	5.09	19.3	26.9
Time of day	1× learned	310.3	22.4	5.90	44.4	13.6
Always update	–	359.8	14.7	2.73	48.8	100.0

Reward-component scales are also computed from the pooled training dataset. For raw reward component j , the shared scale is

$$s_j = \frac{1}{N} \sum_{t=1}^N |r_{t,j}^{\text{raw}}|. \tag{11}$$

A single set of observation and reward statistics is used for all expert and exploration profiles. These statistics are fixed during offline optimization and subsequent policy evaluation.

B Reward Weights

After division by its mean absolute value over the offline dataset, each reward component is multiplied by the following weight: thermal comfort 3; perimeter visual under-illumination 1.5; HVAC electric demand 1; core under-illumination 1; lighting energy 0.4; controllable perimeter over-illumination 0.4; core over-illumination 0.4; artificial-light excess 0.25; daylight over-illumination 0.2; solar-gain penalty while cooling 0.2; solar-gain bonus while heating 0.1; update penalty 0.1. Before normalization, the update penalty charges 0.05 per glazing, 0.01 per lighting and 0.002 per setpoint update. Components are clipped to ± 10 (± 50 for thermal comfort), summed and passed through symlog.

C Update-Rate Sweep

Table S3 lists the values plotted in Figure 2 of the main paper (Section 6.6).

Table S4: Observation-history ablation: our agent with the 36-step history ($K=36$, reported checkpoint) and the same method trained with $K=1$ (the GRU sees only the current observation; one training run). Means over selector seeds ($\pm$ half-range). Δ : $K=1$ minus $K=36$, paired by day, 95% moving-block bootstrap interval (30-day blocks).

History	kWh/d	Viol. %	Kzh/d	Vis. %	Upd. %
$K=36$ (ours) ($n=5$)	344.1 $\pm$ 3.2	7.43 $\pm$ 1.06	1.23 $\pm$ 0.29	48.4 $\pm$ 1.1	13.7 $\pm$ 0.1
$K=1$ ($n=3$)	344.6 $\pm$ 1.1	4.89 $\pm$ 0.04	0.54 $\pm$ 0.01	46.1 $\pm$ 0.6	15.7 $\pm$ 0.2
Δ	+0.5 [-0.5, +4.6]	-2.54 [-3.23, -1.74]	-0.68 [-1.00, -0.28]		

D Observation History

Table S4 compares the reported agent ($K=36$) with the same method trained with a one-step history ($K=1$; one training run, three selector seeds). Electricity is unchanged, and thermal violations and

their severity are lower without the history. Because this is a single training run and the violation rate of our agent varies by about 1.7 pp (SD) across its five training runs, we read it as evidence that the history is not needed in this building rather than as an improvement.

E Baseline Training Length

Stopping the three original baselines of Table 2 of the main paper at 50 instead of 100 epochs does not help: their first training runs then have 27.4% (full-action IQL), 29.6% (full-action BC) and 29.5% (act/repeat BC) occupied thermal violations, against 26.0%, 29.8% and 31.1% after 100 epochs (means over three training runs). The gap to our agent is therefore not caused by training the baselines too long.